

Homography-Guided Multi-Image Super-Resolution

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Abstract—This paper presents a robust pipeline for Multi-Image Super-Resolution (MISR), specifically adapted from the TRMISR architecture to reconstruct diverse surface textures with high structural fidelity. The pipeline features a specialized dataset alignment framework designed to ensure precise spatial registration before upsampling. A core contribution of this work is the creation and utilization of a large-scale synthetic dataset to enhance model generalization. Training leverages a pre-training dataset of 2 656 textures to generate 146 000 synthetic physically-based rendering (PBR) texture patches, followed by fine-tuning on a real-world dataset of 56 distinct scenes containing over 300 000 images. Furthermore, we explore various loss functions to optimize both image sharpness and textural faithfulness, including experiments with Generative Adversarial Networks (GANs) that yield compelling visual results. Quantitative evaluation demonstrates the effectiveness of the proposed approach, achieving an average Peak Signal-to-Noise Ratio (PSNR) of 31.73 dB on the synthetic data and 25.79 dB on the challenging real-world dataset.

Index Terms—Image Super-Resolution, Multi-Image Reconstruction, Homography, Deep Learning, Computer Vision

I. INTRODUCTION

Multi-Image Super-Resolution (MISR) reconstructs high-frequency details by fusing multiple low-resolution (LR) observations of a scene, exploiting natural sub-pixel spatial shifts. Unlike Single-Image Super-Resolution (SISR), which relies on learned priors to “hallucinate” missing details, MISR genuinely recovers lost structural information. However, this approach depends entirely on precise geometric registration. If input frames are not perfectly aligned, the resulting fusion suffers from severe ghosting and blurring, a challenge heavily amplified in real-world captures by complex lens distortions and shifting specular highlights.

To address this, we present an end-to-end MISR pipeline engineered for precise structural fidelity. We implement a robust two-stage alignment framework: utilizing SIFT-based homography for global perspective correction, refined by RAFT optical flow for dense, pixel-level adjustments. For reconstruction, we employ a Transformer-based architecture that leverages multi-head self-attention to effectively aggregate features across the aligned image sequences.

We validate our methodology on two distinct datasets: a perfectly aligned synthetic dataset of 146,000 Physically Based Rendering (PBR) images, and a real-world dataset of over 265,000 smartphone captures. This dual approach allows us to isolate alignment errors from network capacity and analyze the performance drop caused by real-world physical realities, such as non-Lambertian surfaces.

The main objectives of this work are to:

- **Develop an end-to-end MISR architecture** utilizing Transformer-based feature fusion for disparate camera views.
- **Evaluate reconstruction performance** using PSNR, SSIM and LPIPS.
- **Analyze the generalization gap** between idealized synthetic training data and real-world inference.

The remainder of this report is structured as follows: Section II reviews related work; Section III defines the multi-image super-resolution problem and outlines the input alignment constraints adopted in our implementation; Section IV details the methodology; Section V is about data preparation; Section VI describes the experiments and results; and Section VII concludes.

II. RELATED WORK

A significant breakthrough in feature fusion for Multi-Image Super-Resolution was introduced by An *et al.* [1] with TR-MISR. Originally designed for satellite imagery, TR-MISR pioneered the use of Transformer architectures to aggregate information across multiple views with weak temporal correlations. The framework employs a residual encoder to extract features, which are then flattened into vectors. Instead of relying on complex spatial-temporal convolutions, a Transformer module applies dynamic attention to corresponding regions across the image sequence. By utilizing a shared learnable embedding vector to aggregate these multi-view features before decoding via sub-pixel convolution, TR-MISR effectively restores high-frequency details. Our project directly adapts this vector-based Transformer fusion strategy, optimizing it to work in tandem with our explicit geometric alignment pipeline.

III. IMAGE SUPER-RESOLUTION

This section will describe the task of multi image super-resolution and specific constraints applied in our implementation.

A. Problem Definition

Given a set of lower-resolution images (LR) of a planar surface, the task is to reconstruct one of these images (known as *reference*) in a high resolution (HR). The multi-image aspect specifically assumes that capturing a scene from multiple angles provides more information that can be used in reconstructing an image.

B. Input Images Alignment

In our work, we expect the LR inputs to be pre-aligned to the reference image and we focus on reconstructing various planar surface textures.

IV. METHODOLOGY

This section will introduce our model and how it works. First, we will describe the structure of the model, followed by the loss functions and evaluation metrics we used.

A. Architecture Overview

We follow the transformer-based architecture introduced by [1]. It consists of three distinct parts, the encoder, fusion module and decoder. The input to the model is a set of k LR images $\{LR_i\}_1^k$, where LR_1 is the reference to be upsampled. The network architecture overview is shown in Figure 1

1) *Encoder*: We use a HighRes-net encoder, which consists of a convolution layer followed by residual blocks. To highlight differences across input LR images, we calculate an implicit registration image LR_{ref} using the *Median* function, which is then concatenated to each image in the LR sequence.

The encoder transforms each input LR into feature space, formally $Encoder : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^{H \times W \times C_h}$, where C denotes number of color channels, C_h is the hidden feature dimension and H, W are spatial dimensions of the image.

2) *Fusion*: Given a set of LR images processed by the encoder, $\{LR_i\}_1^k$, where $LR_i \in \mathbb{R}^{H \times W \times C_h}$, we concatenate features across the k LR images to obtain $\{x_i^{(h,w)}\}_1^k$ for each h, w coordinate. This set of features is prepended with a learned embedding x_0 , which is shared across all coordinates. This leaves us with a final feature $\{x_i^{(h,w)}\}_0^k$, which is used as the input into the fusion module.

The fusion module consists of classic Transformer encoder blocks, each containing multi-head self-attention layers followed by a feed-forward network. The self-attention is applied along the feature dimension, meaning that for each pixel coordinate (h, w) , we obtain a fused feature vector $f_{(h,w)} \in \mathbb{R}^{C_h}$. Additionally, if a padding mask is provided, the model excludes the all masked coordinates from attention computation. This allows us to restrict the influence of non-informative pixels coming from homography alignments.

3) *Decoder*: To transform the features $f_{(h,w)}$ back to pixel-space and obtain the final super-resolution, we employ 2 convolution layers followed by a subpixel convolution layer, also known as pixel-shuffle [2]. The first convolution layer allows the network to aggregate information from neighborhood features, potentially resulting in smoother transitions between patches of pixels in the resulting super-resolution. This addition was motivated by sharp square artifacts produced by early versions of our network. Second convolution layer follows to resize the feature vector dimensions from C_h to $r^2 C$, where r is the upscale factor and C is the original number of channels of the images. Finally, the subpixel convolution layer reorders the values in the features to obtain the final upscaled image. To mitigate checkerboard artifacts introduced by sub-pixel convolution layers, we apply ICNR (Initialized Convolution

to be Nearest-neighbor Resize) weight initialization [3] to the convolutional layer preceding each PixelShuffle operation.

B. Loss Functions

The choice of the loss function proved to be crucial for the produced image quality. Standard L_1 and L_2 loss functions provide a reasonable baseline; however, training exclusively with either of them results in a network that struggles to reconstruct high-frequency details and sharp edges. The Huber loss, also known as the smooth L_1 loss, combines the properties of both L_1 and L_2 losses by behaving quadratically for small residuals and linearly for large ones, with the transition controlled by a threshold parameter δ :

$$\mathcal{L}_\delta(r) = \begin{cases} \frac{1}{2}r^2 & \text{if } |r| \leq \delta \\ \delta (|r| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

where $r = \hat{I} - I$ denotes the per-pixel residual. This formulation reduces the sensitivity to outliers compared to L_2 loss, while remaining differentiable at zero, unlike L_1 loss.

The Sobel loss penalises deviations in estimated edge gradients between the predicted image $\hat{I}$ and the reference image I . First, both images are convolved with the 3×3 Sobel kernels

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}.$$

The gradient magnitudes of the predicted and reference images are then computed as

$$\nabla \hat{I} = \sqrt{(G_x * \hat{I})^2 + (G_y * \hat{I})^2},$$

$$\nabla I = \sqrt{(G_x * I)^2 + (G_y * I)^2}.$$

Finally, the Sobel loss is defined as

$$\mathcal{L}_{\text{Sobel}} = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W |\nabla \hat{I}_{h,w} - \nabla I_{h,w}|.$$

SSIM (Structural Similarity Index) loss measures perceptual similarity between the images by comparing luminance, contrast, and structure in small local windows. Given two image patches x and y , the SSIM index is defined as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)},$$

where μ_x and μ_y denote the mean intensities of the patches, σ_x^2 and σ_y^2 are the corresponding variances, and σ_{xy} is the covariance between the two patches. The constants C_1 and C_2 are introduced to stabilise the division for small denominator values.

The SSIM loss is then defined as

$$\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(x, y).$$

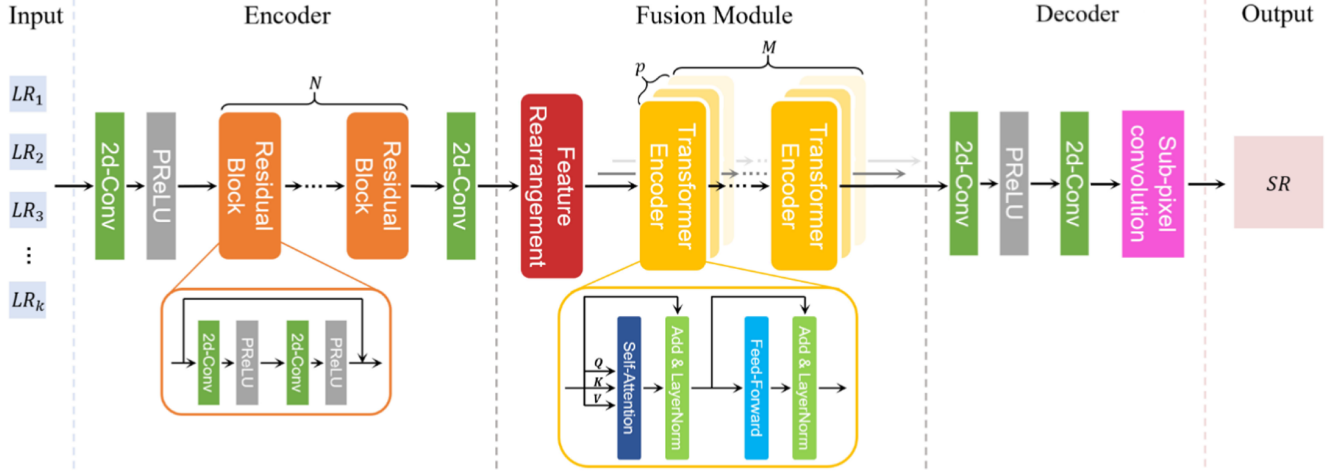


Fig. 1. Our model architecture heavily inspired by TR-MISR model [1]. k denotes the number of input images, N the residual blocks, p the transformer attention heads, and M the transformer layers. Feature rearrangement groups corresponding regions from different images into sequences for transformer processing.

A combination of the Huber, SSIM and Sobel loss can be used to achieve both structural and color aware reconstruction. Therefore, we define a weighted sum of these three functions and refer to it as *combined structural loss*:

$$\mathcal{L}_{\text{STRUCT}} = \alpha \cdot \mathcal{L}_{\delta} + \beta \cdot \mathcal{L}_{\text{SSIM}} + \gamma \mathcal{L}_{\text{Sobel}}, \quad (1)$$

where α , β and γ are individual loss weights.

While the different parts of this structural loss look at pixel-level details to check similarity with the ground-truth image, these metrics do not always match how humans actually perceive quality. As a result, relying only on these pixel-level comparisons might not truly reflect how good the model's predictions are.

Learned Perceptual Image Patch Similarity (LPIPS) evaluates image similarity by comparing feature maps extracted from a pre-trained neural network [4]. Because the entire metric is differentiable, gradients can be backpropagated through the network back to the generative model, making LPIPS highly suitable as a perceptual loss function. We use this to construct a second combined loss function, the *combined perceptual loss*:

$$\mathcal{L}_{\text{PERC}} = \alpha \cdot \mathcal{L}_{\delta} + \beta \cdot \mathcal{L}_{\text{SSIM}} + \gamma \mathcal{L}_{\text{LPIPS}}, \quad (2)$$

where $\mathcal{L}_{\delta}$ the Huber loss, $\mathcal{L}_{\text{SSIM}}$ is the structural similarity index loss, and $\mathcal{L}_{\text{LPIPS}}$ denotes the perceptual distance computed via the deep feature extractor.

C. Generative-Adversarial Network

In our experiments, we observed tendency of our models to produce overly smooth results compared to the ground-truth, even when incorporating structural and perceptual loss functions into the training. Therefore, we experimented with

integrating our model into a Generative-Adversarial Network (GAN) style training. Specifically, we used a VGG-style network architecture described in ESRGAN [5] as a discriminator model.

D. Evaluation Metrics

To objectively assess reconstruction quality, we utilize a combination of pixel-fidelity, structural-integrity, and perceptual metrics. While our framework is capable of multi-channel analysis, we prioritize evaluation on the luminance (Y) channel of the YCbCr color space. This ensures that our quantitative results reflect the structural and contrast accuracy most relevant to human perception, rather than being skewed by chrominance noise.

PSNR (Peak Signal-to-Noise Ratio) serves as our primary metric for measuring pixel-level fidelity. By calculating the ratio between the maximum possible power of the signal and the power of corrupting noise, it provides a rigorous mathematical measure of reconstruction accuracy. We compute PSNR on the luminance channel, as it focuses the evaluation on the high-frequency structural details the network aims to restore.

SSIM (Structural Similarity Index Measure) [6] is used as our secondary metric, evaluating local patterns of luminance, contrast, and structure. Unlike PSNR, which measures absolute pixel differences, SSIM rewards the preservation of edges and textures. It should be noted that SSIM is highly sensitive to spatial alignment; even small registration residuals from the homography process can cause the score to drop if the predicted structures do not perfectly overlap with the ground truth.

Finally, **LPIPS** is employed to capture high-level perceptual quality.

V. DATA PREPARATION

Our models were trained and evaluated using two distinct datasets. First, a large-scale synthetic dataset was constructed using Physically Based Rendering (PBR) textures to simulate high-fidelity images of various surfaces under controlled configurations. This synthetic data was supplemented by a smaller, real-world dataset comprising surface images captured across diverse, unconstrained environments.

A. Synthetic Dataset

TABLE I
SYNTHETIC DATASET STATISTICS

Metric	Value
Number of textures	2 745
Number of scenes	13 725
Image resolution	256×256
Average images per scene	11.00
Total number of image files	150 975

The synthetic dataset was generated using Blender. The virtual scene consists of a standard 3D plane mapped with PBR textures, sourced from the MatSynth dataset [7], along with their corresponding displacement, normal, and auxiliary maps. Illumination is provided by a directional area light source, and data is captured using a virtual camera positioned above the plane. To ensure data diversity, the camera angles, light positions, and light intensities are randomized within predefined operational ranges.

The generation pipeline is structured hierarchically around "scenes," where a single scene represents a unique, randomized lighting configuration. Within each scene, the camera is permuted across multiple viewpoints to capture the texture from various angles. Each asset utilizes multiple scenes, with the exact distribution of textures and scenes summarized in Table I.

The synthetic dataset consists of high-resolution (HR) images rendered at 256×256 pixels, which are downsampled to 64×64 pixels via bicubic interpolation to produce the low-resolution (LR) inputs.

Alignment of the LR frames within a scene is performed using the ground-truth camera matrices obtained directly from the rendering engine. Because the exact camera locations are known, we can directly warp any arbitrary viewpoint to align with the chosen reference frame. During this warping process, binary validity masks are generated to track regions that fall outside the field of view.

To maximize the utility of the dataset, we exploit the multi-view nature of each scene. For a single scene containing N images, we generate up to N distinct training samples by cyclically permuting which image serves as the reference frame.

Augmentation and Degradation: To bridge the gap between synthetic and real-world data, we apply a series of stochastic augmentations on the synthetic data after the alignment step. These include:

- **Gaussian Blur:** To simulate the varying optical characteristics of physical lenses.
- **Gaussian Noise:** To mimic the sensor noise typical of low-light or small-sensor captures.

As a final preprocessing step, we handle the undefined regions identified by the validity masks using a clamp-to-edge technique. By replicating the nearest valid edge pixels to fill the empty areas, we ensure a continuous signal and prevent the model from being distracted by sharp border artifacts.

B. Real-world Dataset

TABLE II
STATISTICS OF THE TILED REAL-WORLD DATASET

Metric	Value
Number of scenes	56
Number of sub-scenes	25 336
Image resolution	256×256
Average images per scene	11.95
Total number of image files	302 815

TABLE III
STATISTICS OF THE REAL-WORLD DATASET

Metric	Value
Total scenes	73
Image resolution	2992×2992
Average images per scene	53.88
Total number of image files	3 933

The real-world dataset, provided by Ing. Michal Hradiš, Ph.D., consists of surface images captured using a standard phone camera. The dataset covers a variety of distinct material types, including gravel, tiles, walls, and other surfaces. Because these images were acquired under unconstrained conditions, they frequently feature real-world illumination challenges, such as harsh shadows and strong specular highlights. The statistics of the original real-world dataset are summarized in Table III. In each scene, images are pre-aligned using a specialized pipeline.¹ The pipeline uses SIFT features to estimate an initial homography transformation, followed by the RAFT model to correct any remaining fine-grained misalignments via optical flow.

After alignment, the images are cut into smaller 256×256 patches. This tiling process dramatically increases the total number of training samples and organizes the data into "sub-scenes." A subscene is simply a collection of tiles taken from the exact same spatial coordinates across all the aligned images in a scene. Each subscene provides the exact multi-frame input sequence required to train our MISR models. The final distribution of data after tiling is summed up in Table II.

VI. EXPERIMENTS

A. Experimental Settings

To evaluate different optimization objectives, training progressed through three phases: pretraining, synthetic variation training, and real-world finetuning. First, the base model was pretrained on clean synthetic data using a standard Huber loss. Next, the model underwent structural variation training on synthetic data using the loss function $\mathcal{L}_{STRUCT}$.

Finally, this model trained with structural loss was finetuned on a real-world dataset under three distinct configurations to analyze performance variations. The first variant (*struct**) maintained the baseline structural loss settings. The second (*perc**) shifted the optimization focus to a perceptual loss function ($\mathcal{L}_{PERC}$). The final variant (*gan*) integrated a generative adversarial framework, pairing a GAN discriminator with an anchor loss that retained the original upstream structural weights. Specific loss formulations and weight configurations for each variant are detailed in Table IV.

General Training Configuration

All variations were trained using the AdamW optimizer with a global effective batch size of 80, distributed across a hardware configuration of four NVIDIA RTX A4000 GPUs (20 samples per GPU, 16 GB VRAM each). Each training sample consisted of 10 low-resolution input frames to generate a $4\times$ upscaled output. To enhance robustness against misaligned inputs, an alignment corruption coefficient of 1.5 was incorporated.

Because the learning rate scheduled varied across both modules and training phases, the exact per-layer rates are organized in Table V.

B. Results

Qualitative Analysis on Synthetic Data: The qualitative results on synthetic data demonstrate that the proposed

transformer-based MISR framework is capable of reconstructing fine geometric structures and repeating textures even under a challenging $4\times$ upscaling setting. As illustrated in Figure 4, the model trained with the structural combination loss $\mathcal{L}_{STRUCT}$ successfully reconstructs high-frequency structures that are difficult to distinguish in individual LR observations. In the first example, the network accurately recovers a fine grid pattern, indicating that the model effectively aggregates sub-pixel information across the input sequence.

The second and third examples in Figure 4 further highlight the model’s ability to reconstruct structured architectural surfaces such as stone pavements and brick cladding. The generated SR outputs preserve sharp edges and consistent texture orientation, while the error maps show that most reconstruction inaccuracies are concentrated around the finest details and high-frequency boundaries that are heavily degraded in the LR inputs.

Figure 2 compares outputs from all evaluated model variants on the same synthetic sample. Although the individual loss formulations produce slightly different reconstruction characteristics, visually, the results are very similar, suggesting that all approaches were able to learn a high quality reconstruction ability. From the metrics perspective, models optimized primarily for pixel-wise fidelity, i.e. those trained with $\mathcal{L}_{STRUCT}$ or Huber loss, have strong PSNR and SSIM performance. The model finetuned on the real-world dataset with $\mathcal{L}_{PERC}$ loss achieves much lower LPIPS score, also aligning with the training objective. Unfortunately, this advantage doesn’t seem to translate back to the synthetic dataset, as can be seen from the metrics results.

The most visually noticeable difference can be spotted in the images produced by the GAN-based model when compared to the others. We observe significantly sharper and visually more detailed textures, achieving the best LPIPS score on the synthetic dataset (Table VI) while retaining comparable PSNR and SSIM scores.

Qualitative Analysis on Real-World Data: The qualitative comparison on real-world captured images is shown in Figure 3. Compared to the synthetic dataset, reconstruction on real

TABLE IV
OVERVIEW OF THE TRAINED MODEL VARIANTS, INCLUDING THEIR TRAINING PHASE, DATASET DOMAIN, OPTIMIZED LOSS FUNCTIONS, AND SPECIFIC COEFFICIENT WEIGHTS.

Model Name	Phase	Dataset Domain	Loss Function	Weight Coefficients
huber-only	Pretraining	Synthetic	$\mathcal{L}_\delta$	Weight = 1.0, $\delta = 0.1$
struct	Variation Training	Synthetic	$\mathcal{L}_{STRUCT}$	$\alpha = 3.5, \beta = 0.6, \gamma = 0.1$
struct*	Finetuning	Real-world	$\mathcal{L}_{STRUCT}$	$\alpha = 3.5, \beta = 0.6, \gamma = 0.1$
perc*	Finetuning	Real-world	$\mathcal{L}_{PERC}$	$\alpha = 1.0, \beta = 0.1, \gamma = 0.1$
gan	Adversarial Training	Real-world	$\mathcal{L}_{STRUCT} + \mathcal{L}_{GAN}$	$1.0 \cdot \mathcal{L}_{STRUCT} + 0.05 \cdot \mathcal{L}_{GAN}$

* Indicates models finetuned on real-world data.

TABLE V
LEARNING RATE CONFIGURATIONS ACROSS DIFFERENT ARCHITECTURAL MODULES AND TRAINING PHASES.

Phase	Encoder	Transformer Fusion	Decoder	Final Proj. Conv
Pretraining	1×10^{-3}	5×10^{-4}	1×10^{-3}	Same as Decoder
Variation and Finetuning	1×10^{-5}	5×10^{-5}	1×10^{-4}	Same as Decoder
Adversarial (GAN)	1×10^{-5}	5×10^{-5}	1×10^{-4}	1×10^{-6}

¹Image alignment scripts: https://github.com/michal-hradis/mess_identification/

TABLE VI
COMPARISON OF IMAGE QUALITY METRICS BY RUN/LOSS FUNCTION TYPE ON REAL AND SYNTHETIC DATASETS.

Method	Real-World Data			Synthetic Data		
	LPIPS ($\downarrow$)	SSIM ($\uparrow$)	PSNR ($\uparrow$)	LPIPS ($\downarrow$)	SSIM ($\uparrow$)	PSNR ($\uparrow$)
Bicubic Interpolation	0.3421	0.8411	25.15	0.2594	0.8399	29.47
gan	0.2569	0.7715	25.68	0.0716	0.8789	30.96
perc*	0.1323	0.8647	25.56	0.1313	0.8818	31.31
struct*	0.1627	0.8713	26.50	0.1395	0.8820	31.43
struct	0.1538	0.8346	23.91	0.1147	0.8901	31.73
huber-only	0.1631	0.8249	23.36	0.1250	0.8858	31.57

* Indicates models finetuned on real-world data.



Fig. 2. Comparison of SR outputs from all evaluated model variants on a synthetic sample. Pixel-wise fidelity models (labeled *Struct* and *Struct**) yield structurally consistent reconstructions with strong PSNR and SSIM scores, while the GAN-based model produces the sharpest textures and achieves the best LPIPS score.

data proved significantly more challenging due to illumination inconsistencies, and the presence of highly detailed irregular textures such as gravel and thin vegetation structures.

The GAN-based model, which performed strongly on synthetic data, performed worse on the real-world dataset. Although the generated outputs often appear visually sharp, they contain noticeable artifacts and inconsistent high-frequency details, leading to substantially lower PSNR and SSIM scores and even weaker LPIPS performance (Table VI). We observed that most of the artifacts appeared in high-frequency detail portions of images, where the model likely struggled the most to infer reliable information about the structure.

In contrast, the fine-tuned variants produced more stable and structurally consistent reconstructions. While their outputs remain slightly blurry compared to the sharpest GAN predictions, they preserve scene geometry more reliably and achieve substantially better quantitative performance.

Challenges with Non-Lambertian Surfaces: The model’s performance significantly degrades when encountering metal-

lic or reflective surfaces, as shown in Figure 5. In these scenarios, the assumption of brightness constancy is violated; capturing the scene from different angles leads to shifting specular highlights and varying reflections across the input sequence. While the model maintains a degree of edge sharpness, it struggles to resolve the inconsistent photometric information. This results in an SR output that lacks color and appears darker than the ground truth. Because the metallic glare changes from one image to the next, the model struggles to merge the conflicting brightness levels into a clear result.

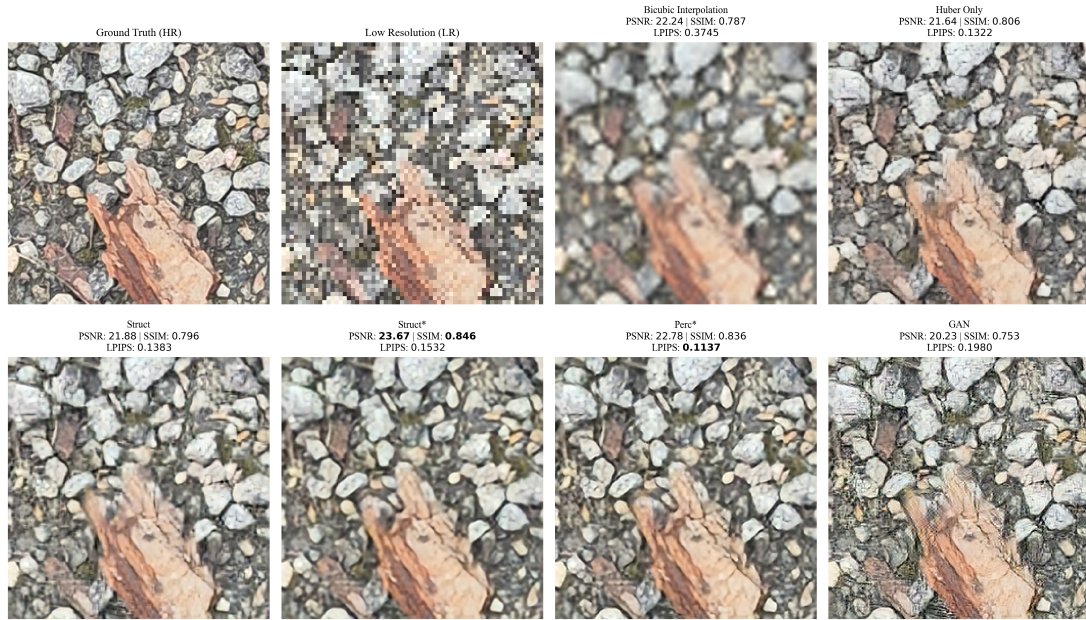


Fig. 3. Comparison of SR outputs from all evaluated model variants on a real-world sample. The GAN-based model, despite its strong performance on the synthetic dataset, produces sharper but artifact-prone outputs that score poorly across all metrics. Fine-tuned variants (marked with “*”) deliver better quantitative results without significant artifacts at the cost of blurry visuals.

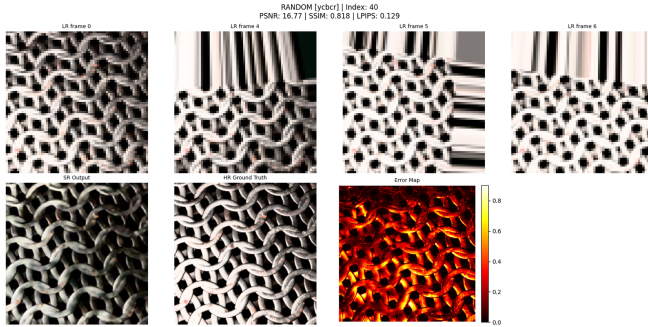


Fig. 5. Analysis of reconstruction failure on metallic surfaces. The top row illustrates how specular highlights shift across the LR images. The resulting SR output demonstrates preserved structural edges but significant luminance and color degradation compared to the HR ground truth.

VII. CONCLUSION

In this work, we presented a Multi-Image Super-Resolution (MISR) framework optimized for structural fidelity on planar surfaces. Our main contribution is the development of several neural networks trained with combinations of structural and perceptual loss functions, as well as with a GAN discriminator. Furthermore, we have developed a pipeline for generating a sizeable high quality synthetic dataset to facilitate the training.

Our experimental evaluation across two distinct datasets highlighted the difficulty of balancing pixel-level faithfulness to the reference with producing a visually pleasing and plausible reconstructions.

We emphasize that the choice of loss functions for training is decisive for the final result, and found out that combining multiple objective functions to cover different aspects of image

quality measurements yields sharper and more faithful results with less artifacts.

A GAN-like training using discriminator network from the ESRGAN [5] paper substantially improved the results on the synthetic dataset, but we failed to replicate this success on the real-world dataset, in which the model produced noticeable artifacts. Further research in this area seems to be promising, as this approach demonstrated the possibility of producing sharper results while retaining faithfulness to the inputs.

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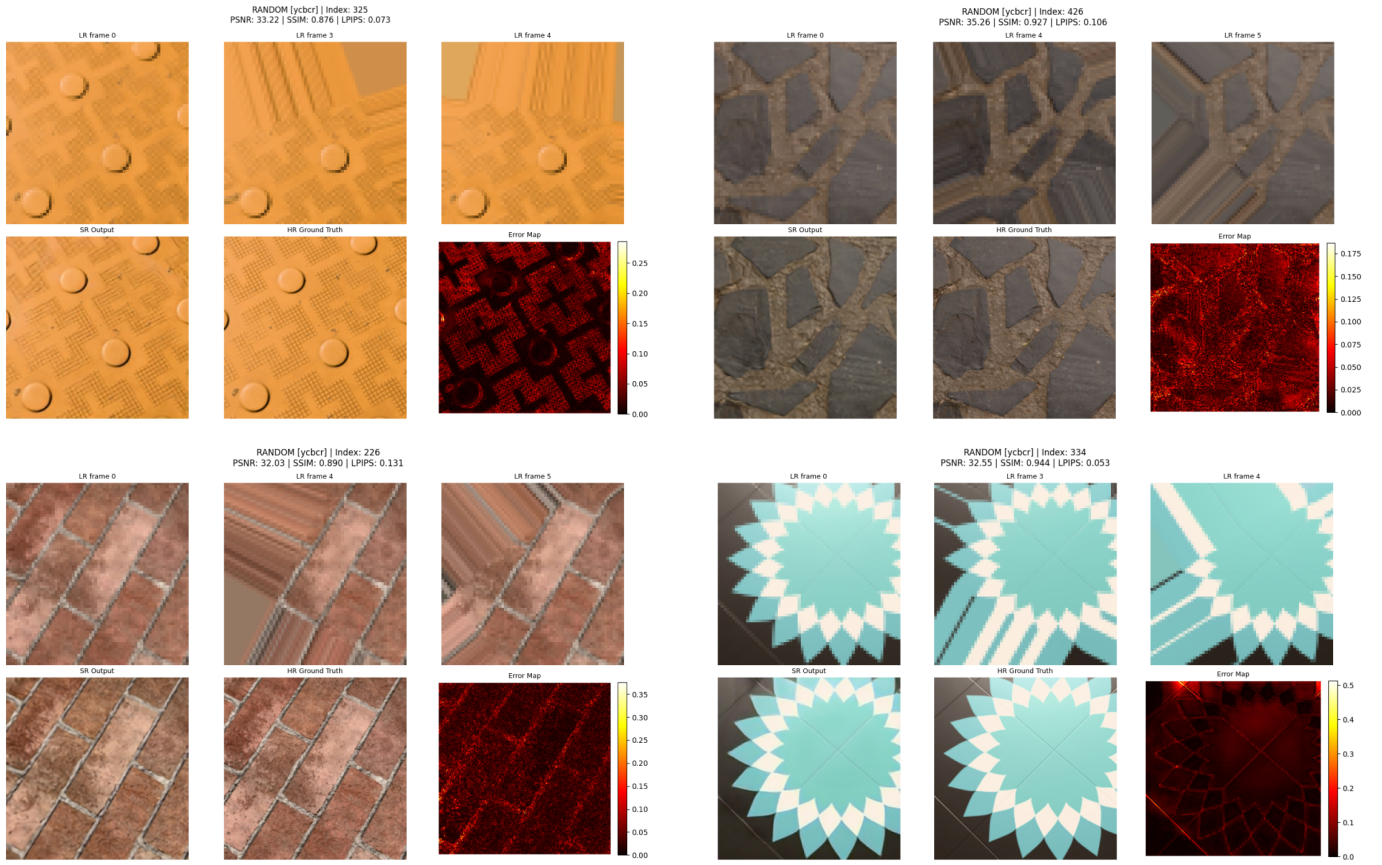


Fig. 4. Qualitative results on synthetic samples. Top: subset of LR input frames. Bottom: comparison between SR output, HR ground truth, and the pixel-wise Error Map. Brighter regions in the error map indicate discrepancies, typically found near high-frequency edges.

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